MAKILESH M

AI Engineer | LLMs • RAG • Voice AI

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PROFESSIONAL SUMMARY

Oracle-certified AI/ML Engineer with hands-on experience building **LLM applications, RAG pipelines, multi-agent systems**, and **real-time Voice AI solutions**. Proven ability to develop production-oriented AI systems with strong focus on low-latency inference, scalable architectures, and practical deployment across cloud and local environments.

SKILLS

AI & Machine Learning: LLMs, RAG, Multi-Agent Systems, NLP, Transformers, Generative AI, Deep Learning, Model Fine-tuning, MLOps, Prompt Engineering, Vector Embeddings, Agentic AI
Frameworks & Libraries: LangChain, LangGraph, CrewAI, AutoGen, PyTorch, TensorFlow, Scikit-learn, Hugging Face, spaCy, OpenCV, Streamlit, Whisper, Twilio
Cloud & Tools: AWS (EC2, S3, Lambda), Azure OpenAI, Docker, FastAPI, Flask, Git, SQL, MySQL, Pinecone, FAISS, ChromaDB, Weaviate, Milvus, Qdrant, n8n
Languages: Python, C++, C, Java, JavaScript, HTML, CSS, React.js, RESTful APIs

WORK EXPERIENCE

AI Developer Intern | Shamla Tech Solutions (Sep 2025 – Feb 2026)

- Deployed an internal **multi-platform lead scraping pipeline** (Reddit, Discord, LinkedIn) with a 3-stage heuristic prefilter (spam detection, job post classification, provider filtering) that cut LLM API calls by 94.6%, using GPT-4-turbo with an Ollama fallback and batch processing across concurrent AsyncIO scrapers and Playwright/Selenium automation.
- Developed an **AI Voice Calling Agent** (Twilio + Pinecone) enabling fully automated outbound calls with zero manual intervention. Implemented persistent conversation memory via Pinecone vector storage, delivering context-aware dynamic responses based on prior interaction history and validated through response quality and latency benchmarking.
- Built a full-duplex **Cross-Platform Meeting Assistant** with real-time 10-speaker diarization using pyannote.audio and streaming transcription across Zoom, Google Meet, Microsoft Teams, and in-person sessions, featuring wake-word activation for on-demand contextual AI expert responses powered by RAG.

PROJECTS

M&A Diligence RAG Engine (Jan 2026 – Feb 2026) - github.com/Makilesh/ma-diligence-rag-engine

- Built a production-grade M&A due diligence RAG engine using a **7-agent LangGraph StateGraph** to automate reasoning over multi-format data rooms (financial statements, legal contracts, board decks) **featuring hybrid dense + sparse retrieval (BAAI/bge-m3 1024 dim + FastEmbed BM25)** over **Qdrant**, fused via custom **RRF**, **cross-encoder reranking (bge-reranker-v2-m3)**, Gemini via LiteLLM with Ollama/Qwen2.5 local fallback, and a **self-corrective loop** with bounded retries and forced refusal over low-confidence context.
- Streams **1000+ page PDFs** via **PyMuPDF** to prevent **OOM**, then stitches financial tables across page breaks using **pdfplumber**. Stores each table in four parallel forms (narrative, row-by-row, metrics, markdown) under a shared table_id, so any retrieval hit pulls all four siblings, keeping arithmetic deterministic and citation chains traceable. Implements **Hierarchical Parent-Child Context Expansion** (512 to 2048 tokens) at synthesis time without fragmenting retrieval.
- Developed **76 automated tests** covering retrieval correctness, async safety, orchestration logic, and end-to-end validation workflows, achieving 100% pipeline completion across evaluation scenarios.

AI-Powered Voice Engine (Sep 2025 – Nov 2025) - github.com/Makilesh/voice_engine_MVP

- Engineered a production-grade real-time full-duplex voice engine on an open-source audio stack achieving **200–400ms end-to-end latency** with **<150ms barge-in detection** via hybrid RMS energy monitoring and selective STT echo filtering on a lock-protected 20ms monitor thread, streaming transcriptions via configurable Whisper tiers (tiny/base/small) and dual Silero + WebRTC VAD, and sentiment-aware tone modulation in a producer-consumer async architecture.
- Built a hybrid TTS layer on a local GPU **Kokoro-82M** (40-90ms first-byte latency) with optional **Cartesia Sonic-3** WebSocket streaming, GPT-4o-mini and a local Ollama fallback, dynamic audio-queue sizing across 3 latency presets, and per-turn STT/LLM/TTS latency instrumentation sustaining stable performance with **zero OOM failures** under load.

CERTIFICATIONS & ACHIEVEMENTS

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional – 2025
- Oracle Cloud Infrastructure 2025 Certified Data Science Professional – 2025
- IMPACT X - Finalist, Top 3 of 200+ teams, Team Lead & AI Engineer, AI Crime Detection System using YOLOv8 (2024)
- THEERKATHON - Winner, Team Lead, Won product incubation opportunity with Forge (2024)
- THIRAN - Finalist, Top 5 of 500+ teams, Team Lead & AI Engineer, AI Legal Engine for Indian Criminal Law (2025)

EDUCATION

Sri Krishna College of Engineering and Technology, Coimbatore, India
B.E. in Computer Science Engineering | CGPA: 8.0/10 | May 2026